Band-1 Sophie Germain: Weak Infinitude Probe (Collapse 2.27)

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Band-1 Sophie Germain: Weak Form Probe (Collapse 2.27)

Honesty stamp. This note records a **finite** enumeration of Sophie Germain primes up to a feasible box bound $N = 10^7$, cross-checked against OEIS A092816 and the Hardy–Littlewood density heuristic. It does **not** prove there are infinitely many Sophie Germain primes. It does **not** claim infinitely many safeprimes. It does **not** claim a Clay-millennial solution. **GATE-04** is locked out of scope.

> **Full infinitude of Sophie Germain primes remains OPEN.** Finite count + heuristic $\neq$ theorem.

Abstract

Definition. A prime p is a **Sophie Germain prime** if $2p+1$ is also prime; then $q = 2p+1$ is the associated **safeprime**.

Literature. Infinitude is conjectured but unproved. The standard heuristic is

$$SG(x) \sim 2C_2 \frac{x}{(\ln x)^2} \approx 1.32032 \frac{x}{(\ln x)^2},$$

where $C_2 \approx 0.6601618158$ is the twin-prime constant. The largest listed Top20 entry has **388,342** digits: $2618163402417 \cdot 2^{1290000} - 1$ (PrimeGrid / L927, Feb 2016).

Box. Every Sophie Germain prime $p \leq N = 10^7$ was enumerated by sieve: **count** $= 56032$ for $p < 10^7$, matching OEIS A092816 exactly (all decade unit tests PASS). HL estimate at 10^7 is ≈ 50822 ($\sim 10\%$ low — known scale of the heuristic at this height).

Problem (Band-1 collapse item 2.27)

1. State the SG / safeprime definitions without inventing theorems.
2. Cite density conjectures and known counts / records honestly.
3. Enumerate on a certified box window; compare to OEIS + HL.
4. Keep full infinitude, Clay rhetoric, and GATE-04 out of the claim surface.

Literature ledger (2026 web check; not re-proved on box)

Item	Status / value	Source
Infinitude of SG primes	**OPEN** (conjectured)	Wikipedia; t5k Top20 FAQ
Infinitely many safeprimes	**OPEN / not claimed here**	equivalent companion
Density heuristic	$2C_2 x/(\ln x)^2$	Hardy–Littlewood analogue
Counts $p < 10^n$	OEIS A092816 (to $n = 14$)	oeis.org/A092816

Item	Status / value	Source
Largest known (digits)	**388{,}342**	t5k.org Top20 Sophie Germain (p)
Sibling conjectures	Dickson / Schinzel H / Bateman–Horn	standard

OEIS A092816 excerpt (number of Sophie Germain primes less than 10^n):

n	1	2	3	4	5	6	7	8	9	10
count	3	10	37	190	1171	7746	**56032**	423140	3308859	26569515

Largest known (Top20 snapshot). Form $2618163402417 \cdot 2^{1290000} - 1$ (388{,}342 digits; Feb 2016). Later large entries appear at lower digit ranks (e.g. Oct 2025, 144{,}653 digits) — none close infinitude.

Method (box)

1. **Sieve.** Eratosthenes boolean array on $[0, 2N+1]$ with $N = 10^7$.
2. **Enumerate.** Collect primes $p \le N$ with $2p+1$ prime.
3. **Unit tests.** Match OEIS A092816 for $p < 10^k$, $k = 1..7$.
4. **Heuristic.** Partial product for C_2 to primes $\le 5 \cdot 10^4$; compare $2C_2N/(\ln N)^2$.
5. **Label.** Finite probe / method demo — **not** a proof of infinitude.

Results

Box window (certified tonight)

Quantity	Value
N	10^7
Range	primes $p \in [2, N]$
Count $p < 10^7$	**56,032**
Matches OEIS A092816	**Yes** (all $k = 1..7$ PASS)
Largest SG $\le N$	**9,999,653**
Associated safeprime	**19,999,307**
Sieve + enum wall	≈ 0.40 s

Hardy–Littlewood check (heuristic only)

Quantity	Value
C_2 (partial $\le 5 \cdot 10^4$)	≈ 0.6601629
C_2 (literature)	≈ 0.6601618158
Estimate at 10^7	≈ 50822
Actual / estimate	≈ 1.10
Proof of infinitude?	**No**

Sample Sophie Germain primes ($\le N$)

First: $2, 3, 5, 11, 23, 29, 41, 53, 83, 89, 113, 131, 173, 179, 191, 233, 239, 251, 281, 293, \ldots$

Last few $\le N$: $\ldots, 9997709, 9998219, 9998771, 9998797, 9998981, 9999083, 9999161, 9999317, 9999503, 9999653$.

Why Band-1 weak form vs full infinitude

- **Band-1 weak (this probe):** finite certified count; OEIS hygiene; HL scale check; literature record cited.
- **Full infinitude (OPEN):** infinitely many SG primes (equivalently safeprimes) — **not proved here**.
- **Do not claim:** infinitely many safeprimes; Clay; ToE; GATE-04 unlock.

Reproducibility

- Lab: `/workspace/mission-control/experiments/band1/sophie_germain/`
- Run: `python3 sophie_germain_enum.py`
- Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `METHOD.md`, `next_wave.md`

Conclusion

Box count: 56,032 Sophie Germain primes with $p < 10^7$ (OEIS match). **Heuristic:** HL $\sim$ 50,822 at the same height (order-correct, not a proof). **Full infinitude: OPEN**. Infinitely many safeprimes: **not claimed**. No Clay claim. GATE-04 locked.